Git Commands

1. git init : Makes a normal folder into a Git project so Git can save and track your code changes.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git init
Reinitialized existing Git repository in /Users/diptifirke/Desktop/VSEC_Assign/.git/
```

2. git status: Shows the current status of your Git repository, such as which files are new, modified, deleted, or ready to be committed.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

3. git add . : Adds all new and changed files to Git, making them ready to be saved in the next commit.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git add .
diptifirke@Diptis-MacBook-Air VSEC_Assign % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

4. git config --list : Displays your Git configuration settings, such as your username, email, default editor, and other Git settings.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
credential.helper=osxkeychain
user.email=diptifirke03@gmail.com
user.name=Diptifirke
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
remote.origin.url=https://github.com/diptifirke/VSEC_Ass.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
branch.main.vscode-merge-base=origin/main
user.email=diptifirke03@gmail.com
```

5. git config user.email "your_email" : Sets the email address that Git will use to identify you in your commits.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git config user.email "diptifirke03@gmail.com"
diptifirke@Diptis-MacBook-Air VSEC_Assign % git add .
diptifirke@Diptis-MacBook-Air VSEC_Assign % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

6. git remote add origin https://github.com/USERNAME/VSEC_ASS.git :
Connects your local Git project to a GitHub repository so you can push and pull code.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git remote add origin https://github.com/diptifir
ke/VSEC_ASS.git
error: remote origin already exists.
diptifirke@Diptis-MacBook-Air VSEC_Assign % git remote -v
origin  https://github.com/diptifirke/VSEC_Ass.git (fetch)
origin  https://github.com/diptifirke/VSEC_Ass.git (push)
```

7. git commit -m "your Message": Saves the added changes as a commit with the message "uploaded".

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git add .
diptifirke@Diptis-MacBook-Air VSEC_Assign % git commit -m "Uploaded"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

8. git branch -M main : Renames your current branch to main.

9. git push -u origin main : Uploads your project to the GitHub main branch and remembers the connection for future pushes.

10. git push : Uploads your saved changes (commits) to GitHub.

```
diptifirke@Diptis-MacBook-Air VSEC_Assign % git branch -M main
diptifirke@Diptis-MacBook-Air VSEC_Assign % git push -u origin main
Enumerating objects: 1460, done.
Counting objects: 100% (1460/1460), done.
Delta compression using up to 10 threads
Compressing objects: 100% (1372/1372), done.
Writing objects: 100% (1457/1457), 1.88 MiB | 2.62 MiB/s, done.
Total 1457 (delta 190), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (190/190), done.
To https://github.com/diptifirke/VSEC_Ass.git
   90b152e..4655c01  main -> main
branch 'main' set up to track 'origin/main'.
```